

suitable for low volumes of up to 150 boards per day.

Semi-automatic. Loading and unloading of PCBs is done manually in this case. These are suitable for professional electronics products with medium volumes of up to 500 boards per day.

Fully-automatic. These are used for precision and in-line high-volume production of more than 500 boards per day.

Nearly 85 per cent of the defects produced in an SMT production line result primarily from the printing process. Careful selection of the printer is therefore imperative. Major considerations for selection of stencil printers are explained below:

Maximum size. Maximum size of the PCB the printer can handle; length and breadth expressed in millimetres

Minimum size. Minimum size of the PCB the printer can handle; length and breadth expressed in millimetres

PCB thickness. Lower and higher limits of the PCB; specified in millimetres

Stencil size. Maximum and minimum size of the stencil; length and breadth specified in millimetres

Printing accuracy. Specified in millimetres and Sigma limits

Cycle time. Standard printing time under optimum condition specified in seconds

Printing mode. Single- or double-side printing

Cleaning system. Dry/wet/vacuum

Operation system. For the integral computer

Automated SMD pick-and-place machines. SMT component placement systems, commonly called pick-and-place machines, are used for placing surface-mount devices (SMDs) onto a PCB. These are also capable of placing of a broad range of electronic components like capacitors, resistors and integrated circuits onto the PCBs from tapes and trays



Fig. 2: Benchtop model of pick-and-place machine



Fig. 3: Automatic desktop pick-and-place machine



Fig. 4: SMT reflow oven

at a high speed with a high degree of precision.

Benchtop pick-and-place models offer low-cost solutions for low- to medium-volume SMT placement applications. Selection of the machine is normally based on the following parameters:

Placement rate. Capacity of placing a number of components in one hour; specified as components per hour (CPH)

Placement accuracy. Accuracy of placing components at designated place on the PCB; specified in μm and Sigma limits

Component size. Dimensions of length and breadth specified in millimetres

Minimum. Minimum size of

component that can be handled

Maximum. Maximum permissible size of components the machine can pick and place

Minimum component lead pitch.

Distance between two adjacent leads; specified in millimetres

Range of components. Capability of placing a variety of components such as LEDs, ball-grid arrays, quad-flat packages and SMT connectors

Picking modes. Capability of picking parts from tapes, tubes or trays and loose components

Software. Windows based, user-friendly

Reflow ovens. Reflow soldering is a process in which solder paste, which holds components temporarily to their contact pads on the PCB, melts and reflows when subjected to controlled heat, thereby permanently connecting the components to pads. There are two types of reflow ovens: infrared (IR) and convection.

In IR reflow ovens, heating is achieved by using IR ceramic heating elements. When assemblies are moved in a specified direction constantly into the conveyors, radiating elements heat the components and heat is moved to the solder paste.

In convection reflow ovens, the PCB is transported through a conveyor system. It has multiple zones, which can be individually controlled for temperature.

Selection criteria for a desktop model of the reflow oven are given below:

Welding requirement. Able to satisfy all welding specifications of 0201 resistance QFP, SOP, PLCC, BGA and CSP.

Size of oven. Able to consistently reflow product at required speed to meet production needs

Maximum board width. Maximum size of PCB the machine can handle

Heating supply. IR ray or hot-air convection

Temperature-control segments. Maximum number of segments that can be set and controlled for temperature; control can be manual or using a PC